

Afzal M. Harish

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PROFESSIONAL SUMMARY

Fourth year student from the British University in Dubai with a 3.81 GPA, graduating by September 2026. My work centres on the intersection of AI and real-world problems, in particular Cybersecurity, where I specifically leverage machine learning to bridge gaps in threat detection and digital infrastructure. I am particularly interested in how AI can automate proactive defence mechanisms within enterprise security and be used in automation for tasks overall in the real world.

EDUCATION

BSc Computer Science (AI) The British University in Dubai	Sep 2022-Sep 2026 Dubai, United Arab Emirates
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WORK EXPERIENCE

Artificial Intelligence Engineer Intern Microsoft	April 2025-May 2025 Remote
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- Successfully completed a mentored internship focused on Cloud Computing and Artificial Intelligence, backed by Microsoft, Edunet and AICTE.
- Developed an end-to-end Heart Disease Detection System using Machine Learning to assist in early-stage cardiovascular diagnosis.
- Engineered a robust hybrid dataset by synthesizing real-world clinical data with secondary online repositories, ensuring model diversity and reducing bias.
- Leveraged Microsoft Azure services to host and manage the model, gaining hands-on experience in cloud architecture and scalable AI deployment.

Digital Technology & Information Security Intern Dubai Islamic Bank	June 2025 – Sep 2025 Onsite, Dubai
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- Partnered with the Digital/IT team to assist customers with procedural tasks involving the software and to identify and document software bugs as well as UI/UX issues within both mobile and web platforms.
- Observed and analysed backend debugging sessions with the core IT team, gaining insight into error replication, logic-based code fixes, and production-level testing cycles.
- Integrated with the Group Internal Security division to shadow network monitoring workflows and Intrusion Detection Systems (IDS) gaining exposure to the cybersecurity side of bank operations.
- Gained technical exposure to enterprise security protocols, including firewalls, VPNs, and secure access frameworks used to protect high-stakes financial data.

PROJECTS

UrbanSight Analytics: Smart City Demographics & Security | AI Researcher & Developer

- Developed a comprehensive, real-time "AI Profiler" that extracts soft biometrics (age, gender, ethnicity, clothing, skin tone, and hair segmentation) from subjects in unconstrained environments.
- Engineered a multi-model pipeline integrating fine-tuned Xception networks (Gender), U-Net (Hair Segmentation), Vision Transformers (ViT), and ONNX models into a unified real-time detection system.
- Designed the system to address challenges in modern Human-Computer Interaction (HCI) and intelligent surveillance, focusing on semantic profiling for targeted retail analytics and security.

- Authored a forthcoming research paper to be published in **IEEE** alongside a fellow colleague at university; presented findings at the 17th Student Research Conference on Applied Computing (SRC 2026) at Zayed University.

Physio-Vision: Hardened AI Motion-Intelligence Ecosystem | AI Developer and Cybersecurity Analyst

- Engineered a full-stack biomechanical analysis system utilizing MediaPipe 3D pose estimation and custom Keras deep learning models for real-time, HIPAA-compliant patient exercise grading.
- Architected a hardened REST API* on a Raspberry Pi featuring Bcrypt adaptive hashing for salted credential storage and JWT-based session rotation to mitigate token theft and replay attacks.
- Implemented a multi-layered perimeter defence integrating Cloudflare Turnstile for frictionless bot protection, strict Pydantic V2 input validation with regex whitelisting, and per-IP tiered rate limiting to neutralize brute-force attempts.
- Developed a secure cross-platform handshake between a PyQt5 desktop application and a web dashboard, utilizing custom "VIP" request headers and OAuth 2.0 loopback servers to protect sensitive biometric data in transit.
- Optimized system-wide audit logging with rotating file handlers to track authentication events and API health, ensuring continuous monitoring and end-to-end data integrity across the ecosystem.
- Co-created and published an open-source biomechanical dataset on Kaggle, optimizing data structure for researcher accessibility and community use.

TECHNICAL SKILLS

Languages | Python, JavaScript, Java, C/C++, SQL, R, HTML5, CSS3

AI & Computer Vision | MediaPipe, PyTorch, TensorFlow, Keras, ONNX Runtime, OpenCV, Scikit-Learn.

Web & Backend | FastAPI, React, RESTful APIs, Uvicorn, Pydantic v2

Security & Auth | JWT (HS256), Bcrypt (Adaptive Hashing), OAuth 2.0, Rate Limiting (SlowAPI), Input Validation.

Frameworks | React, Django, Spring Boot, Flask, Angular, Ruby on Rails, CLI

Developer Tools | Git, Docker, Jenkins, Visual Studio Code, PyCharm, TravisCI, Eclipse, JIRA

Libraries | Pandas, NumPy, Matplotlib, Bootstrap, TensorFlow, jQuery, Redux.

Methodologies | Iterative Development, Full-Stack Development, Peer-Reviewed Research (IEEE Standards).